THE DERIVED TENSOR FUNCTOR WITH A COMPLEX INDUCES MAPS ON DERIVED HOM AND ON EXT

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-upgrade-tensor-with-complex-derived (source lines 6202-6224); the proof environment begins on the second line after it and is its own proof, closing at source line 6274.

Editorial note. The statement and the proof invoke six author results of the same chapter by label plus one section of another chapter; the same-chapter results are retained as context spans and the generator expands the context transitively (budget 800 lines for this card), while reference targets in other chapters that carry canonical project tags are auto-declared as non-load-bearing source pointers.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the derived tensor functor with a complex induces canonical maps on derived Hom and on Ext groups.

Allowed standard prerequisites (editorial).

- Site, sheaf of commutative R-algebras, complexes of O-modules. A site C with a sheaf of commutative R-algebras O and the derived category D(O) of complexes of O-modules.. It is the setting of the statement (source lines 6204-6206).
- Derived tensor product with a fixed complex. The functor - tensor^L_E K^bullet : D(E,d) to D(O) of the earlier lemma (the author's lemma-tensor-with-complex-derived, retained as context).. It is the functor whose induced maps on Hom and Ext are described (statement, source lines 6208-6211).
- Global derived hom and its cohomology. The global derived hom introduced in the Cohomology on Sites chapter, whose cohomology modules compute Ext over O (declared as a tagged source pointer).. It identifies the right hand side of the arrow (source lines 6226-6229).
- Derived restriction and property (P) computations. The derived Hom realised through the restriction bimodule, invariant under replacing M and N by property-(P) dg E-modules, computed as a dg Hom of bimodules (the author's lemmas lemma-functoriality-derived-restriction and lemma-compute-derived-restriction, and the section section-restriction, retained as

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context).. They identify the left hand side and justify the replacement (source lines 6230-6243).

Source reference. Original lemma 0B6E: sources/stacks-chow/tags.txt, source lines 14067–14067. Cross-chapter pointer whose target carries the canonical tag 0B6E in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 06XN: sources/stacks-chow/tags.txt, source lines 8857–8857. Cross-chapter pointer whose target carries the canonical tag 06XN in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 07KW: sources/stacks-chow/tags.txt, source lines 9671–9671. Cross-chapter pointer whose target carries the canonical tag 07KW in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09IZ: sources/stacks-chow/tags.txt, source lines 12057–12057. Cross-chapter pointer whose target carries the canonical tag 09IZ in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0149: sources/stacks-chow/tags.txt, source lines 1464–1464. Cross-chapter pointer whose target carries the canonical tag 0149 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 09J0: sources/stacks-chow/tags.txt, source lines 12058–12058. Cross-chapter pointer whose target carries the canonical tag 09J0 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0FNG: sources/stacks-chow/tags.txt, source lines 19566–19566. Cross-chapter pointer whose target carries the canonical tag 0FNG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 014C: sources/stacks-chow/tags.txt, source lines 1467–1467. Cross-chapter pointer whose target carries the canonical tag 014C in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0125: sources/stacks-chow/tags.txt, source lines 1387–1387. Cross-chapter pointer whose target carries the canonical tag 0125 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 013N: sources/stacks-chow/tags.txt, source lines 1442–1442. Cross-chapter pointer whose target carries the canonical tag 013N in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *The functor $K(Mod_{(E,d)}) \to K(\mathcal{O})$ of Lemma 0.5 has a left derived version defined on all of $D(E,d)$. We denote it $- \otimes^{\mathbf{L}}_E K^\bullet : D(E,d) \to D(\mathcal{O})$.*

Proof. We will use Derived Categories, Lemma 06XN to prove this. As our collection $\mathcal{P}$ of objects we will use the objects with property (P). Property (1) was shown in Lemma 1.2. Property (2) holds because if $s : P \to P'$ is a quasi-isomorphism of modules with property (P), then s is a homotopy equivalence by Lemma 1.1. □

Lemma 0.2. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let $f : N \to N'$ be a homomorphism of differential graded (A, B)-bimodules. Then f induces a morphism of functors*

$$- \circ f : R\operatorname{Hom}(N', -) \longrightarrow R\operatorname{Hom}(N, -)$$

If f is a quasi-isomorphism, then $f \circ -$ is an isomorphism of functors.

Proof. Write $\mathcal{B} = \operatorname{Mod}^{dg}_{(B,\mathrm{d})}$ the differential graded category of differential graded B-modules, see Example 1.3. Let I be a differential graded B-module with property (I). Then $f \circ - : \operatorname{Hom}_{\mathcal{B}}(N', I) \to \operatorname{Hom}_{\mathcal{B}}(N, I)$ is a map of differential graded A-modules. Moreover, this is functorial with respect to I. Since the functors $R\operatorname{Hom}(N', -)$ and $R\operatorname{Hom}(N, -)$ are computed by applying $\operatorname{Hom}_{\mathcal{B}}$ into objects with property (I) (Lemma 1.4) we obtain a transformation of functors as indicated.

Assume that f is a quasi-isomorphism. Let $F_\bullet$ be the given filtration on I. Since $I = \lim I/F_pI$ we see that $\operatorname{Hom}_{\mathcal{B}}(N', I) = \lim \operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$ and $\operatorname{Hom}_{\mathcal{B}}(N, I) = \lim \operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$. Since the transition maps in the system I/F_pI are split as graded modules, we see that the transition maps in the systems $\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$ and $\operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$ are surjective. Hence $\operatorname{Hom}_{\mathcal{B}}(N', I)$, resp. $\operatorname{Hom}_{\mathcal{B}}(N, I)$ viewed as a complex of abelian groups computes $R\lim$ of the system of complexes $\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI)$, resp. $\operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$. See More on Algebra, Lemma 07KW. Thus it suffices to prove each

$$\operatorname{Hom}_{\mathcal{B}}(N', I/F_pI) \to \operatorname{Hom}_{\mathcal{B}}(N, I/F_pI)$$

is a quasi-isomorphism. Since the surjections $I/F_{p+1}I \to I/F_pI$ are split as maps of graded B-modules we see that

$$0 \to \operatorname{Hom}_{\mathcal{B}}(N', F_pI/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N', I/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N', I/F_pI) \to 0$$

is a short exact sequence of differential graded A-modules. There is a similar sequence for N and f induces a map of short exact sequences. Hence by induction on p (starting with $p = 0$ when $I/F_0I = 0$) we conclude that it suffices to show that the map $\operatorname{Hom}_{\mathcal{B}}(N', F_pI/F_{p+1}I) \to \operatorname{Hom}_{\mathcal{B}}(N, F_pI/F_{p+1}I)$ is a quasi-isomorphism. Since $F_pI/F_{p+1}I$ is a product of shifts of $A^\vee$ it suffice to prove $\operatorname{Hom}_{\mathcal{B}}(N', B^\vee[k]) \to \operatorname{Hom}_{\mathcal{B}}(N, B^\vee[k])$ is a quasi-isomorphism. By Lemma 1.5 it suffices to show $(N')^\vee \to N^\vee$ is a quasi-isomorphism. This is true because f is a quasi-isomorphism and $(\ )^\vee$ is an exact functor. □

Lemma 0.3. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let N be a differential graded (A, B)-bimodule. If $\operatorname{Hom}_{D(B,d)}(N, N') = \operatorname{Hom}_{K(Mod_{(B,d)})}(N, N')$ for all $N' \in K(B, d)$, for example if N has property (P) as a differential graded B-module, then*

$$R\operatorname{Hom}(N, M) = \operatorname{Hom}_{Mod^{dg}_{(B,d)}}(N, M)$$

functorially in M in $D(B, d)$.

Proof. By construction (Lemma 1.4) to find $R\mathrm{Hom}(N, M)$ we choose a quasi-isomorphism $M \to I$ where I is a differential graded B-module with property (I) and we set $R\mathrm{Hom}(N, M) = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, I)$. By assumption the map

$$\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, M) \longrightarrow \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, I)$$

induced by $M \to I$ is a quasi-isomorphism, see discussion in Example 1.3. This proves the lemma. If N has property (P) as a B-module, then we see that the assumption is satisfied by Lemma 1.1. □

Lemma 0.4. *In the situation above there is a functor*

$$- \otimes_E K^\bullet : Mod^{dg}_{(E,d)} \longrightarrow Comp^{dg}(\mathcal{O})$$

of differential graded categories. This functor sends E to $K^\bullet$ and commutes with direct sums.

Proof. Let M be a differential graded E-module. For every object U of $\mathcal{C}$ the complex $K^\bullet(U)$ is a left differential graded E-module as well as a right $\mathcal{O}(U)$-module. The actions commute, so we have a bimodule. Thus, by the constructions in Sections 3 and 2 we can form the tensor product

$$M \otimes_E K^\bullet(U)$$

which is a differential graded $\mathcal{O}(U)$-module, i.e., a complex of $\mathcal{O}(U)$-modules. This construction is functorial with respect to U, hence we can sheafify to get a complex of $\mathcal{O}$-modules which we denote

$$M \otimes_E K^\bullet$$

Moreover, for each U the construction determines a functor $\mathrm{Mod}^{dg}_{(E,\mathrm{d})} \to \mathrm{Comp}^{dg}(\mathcal{O}(U))$ of differential graded categories by Lemma 1.6. It is therefore clear that we obtain a functor as stated in the lemma. □

Lemma 0.5. *The functor of Lemma 0.4 defines an exact functor of triangulated categories $K(Mod_{(E,d)}) \to K(\mathcal{O})$.*

Proof. The functor induces a functor between homotopy categories by Lemma 3.1. We have to show that $- \otimes_E K^\bullet$ transforms distinguished triangles into distinguished triangles. Suppose that $0 \to K \to L \to M \to 0$ is an admissible short exact sequence of differential graded E-modules. Let $s : M \to L$ be a graded E-module homomorphism which is left inverse to $L \to M$. Then s defines a map $M \otimes_E K^\bullet \to L \otimes_E K^\bullet$ of graded $\mathcal{O}$-modules (i.e., respecting $\mathcal{O}$-module structure and grading, but not differentials) which is left inverse to $L \otimes_E K^\bullet \to M \otimes_E K^\bullet$. Thus we see that

$$0 \to K \otimes_E K^\bullet \to L \otimes_E K^\bullet \to M \otimes_E K^\bullet \to 0$$

is a termwise split short exact sequences of complexes, i.e., a defines a distinguished triangle in $K(\mathcal{O})$. □

1. Derived Hom

Lemma 1.1. *Let (A, d) be a differential graded algebra. Let M and N be differential graded A-modules.*

(1) *Let $P \to M$ be a P-resolution as in Lemma 1.2. Then*

$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(P, N)$$

(2) *Let $N \to I$ be an I-resolution as in Lemma 3.4. Then*

$$\mathrm{Hom}_{D(A,d)}(M, N) = \mathrm{Hom}_{K(Mod_{(A,d)})}(M, I)$$

Proof. Let $P \to M$ be as in (1). Since $P \to M$ is a quasi-isomorphism we see that

$$\mathrm{Hom}_{D(A,\mathrm{d})}(P, N) = \mathrm{Hom}_{D(A,\mathrm{d})}(M, N)$$

by definition of the derived category. A morphism $f : P \to N$ in $D(A, \mathrm{d})$ is equal to $s^{-1}f'$ where $f' : P \to N'$ is a morphism and $s : N \to N'$ is a quasi-isomorphism. Choose a distinguished triangle

$$N \to N' \to Q \to N[1]$$

As s is a quasi-isomorphism, we see that Q is acyclic. Thus $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, Q[k]) = 0$ for all k by Lemma 3.3. Since $\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(P, -)$ is cohomological, we conclude that we can lift $f' : P \to N'$ uniquely to a morphism $f : P \to N$. This finishes the proof.

The proof of (2) is dual to that of (1) using Lemma 3.2 in stead of Lemma 3.3. □

Lemma 1.2. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules such that*

(1) *$P \to M$ is a quasi-isomorphism, and*
(2) *P has property (P).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_{i+1} \to P_i \to M_i \to 0$$

where the maps $P_i \to M_i$ are chosen as in Lemma 3.5. This gives a "resolution"

$$\ldots \to P_2 \xrightarrow{f_2} P_1 \xrightarrow{f_1} P_0 \to M \to 0$$

Then we set

$$P = \bigoplus\nolimits_{i \geq 0} P_i$$

as an A-module with grading given by $P^n = \bigoplus_{a+b=n} P^b_{-a}$ and differential (as in the construction of the total complex associated to a double complex) by

$$\mathrm{d}_P(x) = f_{-a}(x) + (-1)^a \mathrm{d}_{P_{-a}}(x)$$

for $x \in P^b_{-a}$. With these conventions P is indeed a differential graded A-module. Recalling that each P_i has a two step filtration $0 \to P'_i \to P_i \to P''_i \to 0$ we set

$$F_{2i}P = \bigoplus\nolimits_{i \geq j \geq 0} P_j \subset \bigoplus\nolimits_{i \geq 0} P_i = P$$

and we add P'_{i+1} to $F_{2i}P$ to get F_{2i+1}. These are differential graded submodules and the successive quotients are direct sums of shifts of A. By Lemma 3.6 we see that the inclusions $F_iP \to F_{i+1}P$ are admissible monomorphisms. Finally, we have to show that the map $P \to M$ (given by the augmentation $P_0 \to M$) is a quasi-isomorphism. This follows from Homology, Lemma 09IZ. □

Example 1.3 (Differential graded category of differential graded modules)**.** Let (A, d) be a differential graded algebra over a ring R. We will construct a differential graded category $\mathrm{Mod}^{dg}_{(A,\mathrm{d})}$ over R whose category of complexes is $\mathrm{Mod}_{(A,\mathrm{d})}$ and whose homotopy category is $K(\mathrm{Mod}_{(A,\mathrm{d})})$. As objects of $\mathrm{Mod}^{dg}_{(A,\mathrm{d})}$ we take the

differential graded A-modules. Given differential graded A-modules L and M we set

$$\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(L, M) = \mathrm{Hom}_{\mathrm{Mod}^{gr}_{A}}(L, M) = \bigoplus \mathrm{Hom}^n(L, M)$$

as a graded R-module where the right hand side is defined as in Example 3.9. In other words, the nth graded piece $\mathrm{Hom}^n(L, M)$ is the R-module of right A-module maps homogeneous of degree n. For an element $f \in \mathrm{Hom}^n(L, M)$ we set

$$\mathrm{d}(f) = \mathrm{d}_M \circ f - (-1)^n f \circ \mathrm{d}_L$$

To make sense of this we think of d_M and d_L as graded R-module maps and we use composition of graded R-module maps. It is clear that $\mathrm{d}(f)$ is homogeneous of degree $n + 1$ as a graded R-module map, and it is A-linear because

$$\begin{aligned}
\mathrm{d}(f)(xa) &= \mathrm{d}_M(f(x)a) - (-1)^n f(\mathrm{d}_L(xa)) \\
&= \mathrm{d}_M(f(x))a + (-1)^{\deg(x)+n} f(x)\mathrm{d}(a) - (-1)^n f(\mathrm{d}_L(x))a - (-1)^{n+\deg(x)} f(x)\mathrm{d}(a) \\
&= \mathrm{d}(f)(x)a
\end{aligned}$$

as desired (observe that this calculation would not work without the sign in the definition of our differential on Hom). Similar formulae to those of Example 3.8 hold for the differential of f in terms of components. The reader checks (in the same way as in Example 3.8) that

(1) d has square zero,
(2) an element f in $\mathrm{Hom}^n(L, M)$ has $\mathrm{d}(f) = 0$ if and only if $f : L \to M[n]$ is a homomorphism of differential graded A-modules,
(3) in particular, the category of complexes of $\mathrm{Mod}^{dg}_{(A,\mathrm{d})}$ is $\mathrm{Mod}_{(A,\mathrm{d})}$,
(4) the homomorphism defined by f as in (2) is homotopy equivalent to zero if and only if $f = \mathrm{d}(g)$ for some $g \in \mathrm{Hom}^{n-1}(L, M)$.
(5) in particular, we obtain a canonical isomorphism

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(L, M) \longrightarrow H^0(\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(L, M))$$

and the homotopy category of $\mathrm{Mod}^{dg}_{(A,\mathrm{d})}$ is $K(\mathrm{Mod}_{(A,\mathrm{d})})$.

Given differential graded A-modules K, L, M we define composition

$$\mathrm{Hom}^m(L, M) \times \mathrm{Hom}^n(K, L) \longrightarrow \mathrm{Hom}^{n+m}(K, M)$$

by composition of homogeneous right A-module maps $(g, f) \mapsto g \circ f$. This defines a map of differential graded modules

$$\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(L, M) \otimes_R \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(K, L) \longrightarrow \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(K, M)$$

as required in Definition 3.7 because

$$\begin{aligned}
\mathrm{d}(g \circ f) &= \mathrm{d}_M \circ g \circ f - (-1)^{n+m} g \circ f \circ \mathrm{d}_K \\
&= (\mathrm{d}_M \circ g - (-1)^m g \circ \mathrm{d}_L) \circ f + (-1)^m g \circ (\mathrm{d}_L \circ f - (-1)^n f \circ \mathrm{d}_K) \\
&= \mathrm{d}(g) \circ f + (-1)^m g \circ \mathrm{d}(f)
\end{aligned}$$

as desired.

Lemma 1.4. *In the situation above, the right derived functor of F exists. We denote it $R\,\mathrm{Hom}(N, -) : D(B, d) \to D(A, d)$.*

Proof. We will use Derived Categories, Lemma 06XN to prove this. As our collection $\mathcal{I}$ of objects we will use the objects with property (I). Property (1) was shown in Lemma 3.4. Property (2) holds because if $s : I \to I'$ is a quasi-isomorphism of modules with property (I), then s is a homotopy equivalence by Lemma 1.1. □

Lemma 1.5. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(M, A^\vee[k]) = \mathrm{Ker}(d : (M^\vee)^k \to (M^\vee)^{k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(M, A^\vee[k]) = H^k(M^\vee)$$

as functors in the differential graded A-module M.

Proof. This is clear from the discussion above. □

Lemma 1.6. *Let R be a ring. Let (A, d) and (B, d) be differential graded algebras over R. Let N be a differential graded (A, B)-bimodule. Then $M \mapsto M \otimes_A N$ defines a functor*

$$- \otimes_A N : Mod^{dg}_{(A,d)} \longrightarrow Mod^{dg}_{(B,d)}$$

of differential graded categories. This functor induces functors

$$Mod_{(A,d)} \to Mod_{(B,d)} \quad \text{and} \quad K(Mod_{(A,d)}) \to K(Mod_{(B,d)})$$

by an application of Lemma 3.1.

Proof. Above we have seen how the construction defines a functor of underlying graded categories. Thus it suffices to show that the construction is compatible with differentials. Let M and M' be differential graded A-modules and let $f : M \to M'$ be an A-module homomorphism which is homogeneous of degree n. Then we have

$$\mathrm{d}(f) = \mathrm{d}_{M'} \circ f - (-1)^n f \circ \mathrm{d}_M$$

On the other hand, we have

$$\mathrm{d}(f \otimes \mathrm{id}_N) = \mathrm{d}_{M' \otimes_A N} \circ (f \otimes \mathrm{id}_N) - (-1)^n (f \otimes \mathrm{id}_N) \circ \mathrm{d}_{M \otimes_A N}$$

Applying this to an element $x \otimes y$ with $x \in M$ and $y \in N$ homogeneous we get

$$\begin{aligned} \mathrm{d}(f \otimes \mathrm{id}_N)(x \otimes y) =& \mathrm{d}_{M'}(f(x)) \otimes y + (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ & - (-1)^n f(\mathrm{d}_M(x)) \otimes y - (-1)^{n+\deg(x)} f(x) \otimes \mathrm{d}_N(y) \\ =& \mathrm{d}(f)(x \otimes y) \end{aligned}$$

Thus we see that $\mathrm{d}(f) \otimes \mathrm{id}_N = \mathrm{d}(f \otimes \mathrm{id}_N)$ and the proof is complete. □

2. Bimodules

3. Tensor product

Lemma 3.1. *Let R be a ring. A functor $F : \mathcal{A} \to \mathcal{B}$ of differential graded categories over R induces functors $Comp(\mathcal{A}) \to Comp(\mathcal{B})$ and $K(\mathcal{A}) \to K(\mathcal{B})$.*

Proof. Omitted. □

Lemma 3.2. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module with property (I). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(N, I) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 3.11). Let $F_\bullet$ be a filtration on I as in property (I). The short exact sequence of Lemma 3.10 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, I/F_iI) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules I/F_iI has a finite filtration by admissible monomorphisms, whose graded pieces are products of $A^\vee[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(N, A^\vee[k]) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 1.5 and the fact that $(-)^\vee$ is an exact functor. □

Lemma 3.3. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module with property (P). Then*

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(P, N) = 0$$

for all acyclic differential graded A-modules N.

Proof. We will use that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a triangulated category (Proposition 3.11). Let $F_\bullet$ be a filtration on P as in property (P). The short exact sequence of Lemma 3.13 produces a distinguished triangle. Hence by Derived Categories, Lemma 0149 it suffices to show that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(F_iP, N) = 0$$

for all acyclic differential graded A-modules N and all i. Each of the differential graded modules F_iP has a finite filtration by admissible monomorphisms, whose graded pieces are direct sums of shifts $A[k]$. Thus it suffices to prove that

$$\mathrm{Hom}_{K(\mathrm{Mod}_{(A,\mathrm{d})})}(A[k], N) = 0$$

for all acyclic differential graded A-modules N and all k. This follows from Lemma 3.12. □

Lemma 3.4. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules such that*

1. *$M \to I$ is a quasi-isomorphism, and*
2. *I has property (I).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_i \to I_i \to M_{i+1} \to 0$$

where the maps $M_i \to I_i$ are chosen as in Lemma 3.14. This gives a "resolution"

$$0 \to M \to I_0 \xrightarrow{f_0} I_1 \xrightarrow{f_1} I_1 \to \ldots$$

Denote I the differential graded A-module with graded parts

$$I^n = \prod\nolimits_{i \geq 0} I_i^{n-i}$$

and differential defined by

$$\mathrm{d}_I(x) = f_i(x) + (-1)^i \mathrm{d}_{I_i}(x)$$

for $x \in I_i^{n-i}$. With these conventions I is indeed a differential graded A-module. Recalling that each I_i has a two step filtration $0 \to I_i' \to I_i \to I_i'' \to 0$ we set

$$F_{2i}I^n = \prod_{j \geq i} I_j^{n-j} \subset \prod_{i \geq 0} I_i^{n-i} = I^n$$

and we add a factor I_{i+1}' to $F_{2i}I$ to get $F_{2i+1}I$. These are differential graded submodules and the successive quotients are products of shifts of $A^\vee$. By Lemma 3.15 we see that the inclusions $F_{i+1}I \to F_iI$ are admissible monomorphisms. Finally, we have to show that the map $M \to I$ (given by the augmentation $M \to I_0$) is a quasi-isomorphism. This follows from Homology, Lemma 09J0. □

Lemma 3.5. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules with the following properties*

1. *$P \to M$ is surjective,*
2. *$\mathrm{Ker}(d_P) \to \mathrm{Ker}(d_M)$ is surjective, and*
3. *P sits in an admissible short exact sequence $0 \to P' \to P \to P'' \to 0$ where P', P'' are direct sums of shifts of A.*

Proof. Let P_k be the free A-module with generators x, y in degrees k and $k+1$. Define the structure of a differential graded A-module on P_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. For every element $m \in M^k$ there is a homomorphism $P_k \to M$ sending x to m and y to $\mathrm{d}(m)$. Thus we see that there is a surjection from a direct sum of copies of P_k to M. This clearly produces $P \to M$ having properties (1) and (3). To obtain property (2) note that if $m \in \mathrm{Ker}(\mathrm{d}_M)$ has degree k, then there is a map $A[k] \to M$ mapping 1 to m. Hence we can achieve (2) by adding a direct sum of copies of shifts of A. □

Lemma 3.6. *Let (A, d) be a differential graded algebra. Let $M \to P$ be a surjective homomorphism of differential graded A-modules. If P is projective as a graded A-module, then $M \to P$ is an admissible epimorphism.*

Proof. This is immediate from the definitions. □

Definition 3.7. Let R be a ring. A *differential graded category $\mathcal{A}$ over R* is a category where every morphism set is given the structure of a differential graded R-module and where for $x, y, z \in \mathrm{Ob}(\mathcal{A})$ composition is R-bilinear and induces a homomorphism

$$\mathrm{Hom}_{\mathcal{A}}(y, z) \otimes_R \mathrm{Hom}_{\mathcal{A}}(x, y) \longrightarrow \mathrm{Hom}_{\mathcal{A}}(x, z)$$

of differential graded R-modules.

Example 3.8 (Differential graded category of complexes)**.** Let $\mathcal{B}$ be an additive category. We will construct a differential graded category $\mathrm{Comp}^{dg}(\mathcal{B})$ over $R = \mathbf{Z}$ whose associated category of complexes is $\mathrm{Comp}(\mathcal{B})$ and whose associated homotopy category is $K(\mathcal{B})$. As objects of $\mathrm{Comp}^{dg}(\mathcal{B})$ we take complexes of $\mathcal{B}$. Given complexes $A^\bullet$ and $B^\bullet$ of $\mathcal{B}$, we sometimes also denote $A^\bullet$ and $B^\bullet$ the corresponding graded objects of $\mathcal{B}$ (i.e., forget about the differential). Using this abuse of notation, we set

$$\mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{B})}(A^\bullet, B^\bullet) = \mathrm{Hom}_{\mathrm{Gr}^{gr}(\mathcal{B})}(A^\bullet, B^\bullet) = \bigoplus\nolimits_{n \in \mathbf{Z}} \mathrm{Hom}^n(A, B)$$

as a graded $\mathbf{Z}$-module with notation and definitions as in Example 3.16. In other words, the nth graded piece is the abelian group of homogeneous morphism of degree n of graded objects

$$\mathrm{Hom}^n(A^\bullet, B^\bullet) = \prod\nolimits_{p+q=n} \mathrm{Hom}_{\mathcal{B}}(A^{-q}, B^p)$$

Observe reversal of indices and observe we have a direct product and not a direct sum. For an element $f \in \mathrm{Hom}^n(A^\bullet, B^\bullet)$ of degree n we set

$$\mathrm{d}(f) = \mathrm{d}_B \circ f - (-1)^n f \circ \mathrm{d}_A$$

The sign is exactly as in More on Algebra, Section 0FNG. To make sense of this we think of d_B and d_A as maps of graded objects of $\mathcal{B}$ homogeneous of degree 1 and we use composition in the category $\mathrm{Gr}^{gr}(\mathcal{B})$ on the right hand side. In terms of components, if $f = (f_{p,q})$ with $f_{p,q} : A^{-q} \to B^p$ we have

$$\mathrm{d}(f_{p,q}) = \mathrm{d}_B \circ f_{p,q} - (-1)^{p+q} f_{p,q} \circ \mathrm{d}_A \tag{3.8.1}$$

Note that the first term of this expression is in $\mathrm{Hom}_{\mathcal{B}}(A^{-q}, B^{p+1})$ and the second term is in $\mathrm{Hom}_{\mathcal{B}}(A^{-q-1}, B^p)$. The reader checks that

(1) d has square zero,
(2) an element f in $\mathrm{Hom}^n(A^\bullet, B^\bullet)$ has $\mathrm{d}(f) = 0$ if and only if the morphism $f : A^\bullet \to B^\bullet[n]$ of graded objects of $\mathcal{B}$ is actually a map of complexes,
(3) in particular, the category of complexes of $\mathrm{Comp}^{dg}(\mathcal{B})$ is equal to $\mathrm{Comp}(\mathcal{B})$,
(4) the morphism of complexes defined by f as in (2) is homotopy equivalent to zero if and only if $f = \mathrm{d}(g)$ for some $g \in \mathrm{Hom}^{n-1}(A^\bullet, B^\bullet)$.
(5) in particular, we obtain a canonical isomorphism

$$\mathrm{Hom}_{K(\mathcal{B})}(A^\bullet, B^\bullet) \longrightarrow H^0(\mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{B})}(A^\bullet, B^\bullet))$$

and the homotopy category of $\mathrm{Comp}^{dg}(\mathcal{B})$ is equal to $K(\mathcal{B})$.

Given complexes $A^\bullet$, $B^\bullet$, $C^\bullet$ we define composition

$$\mathrm{Hom}^m(B^\bullet, C^\bullet) \times \mathrm{Hom}^n(A^\bullet, B^\bullet) \longrightarrow \mathrm{Hom}^{n+m}(A^\bullet, C^\bullet)$$

by composition $(g, f) \mapsto g \circ f$ in the graded category $\mathrm{Gr}^{gr}(\mathcal{B})$, see Example 3.16. This defines a map of differential graded modules

$$\mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{B})}(B^\bullet, C^\bullet) \otimes_R \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{B})}(A^\bullet, B^\bullet) \longrightarrow \mathrm{Hom}_{\mathrm{Comp}^{dg}(\mathcal{B})}(A^\bullet, C^\bullet)$$

as required in Definition 3.7 because

$$\begin{aligned}
\mathrm{d}(g \circ f) &= \mathrm{d}_C \circ g \circ f - (-1)^{n+m} g \circ f \circ \mathrm{d}_A \\
&= (\mathrm{d}_C \circ g - (-1)^m g \circ \mathrm{d}_B) \circ f + (-1)^m g \circ (\mathrm{d}_B \circ f - (-1)^n f \circ \mathrm{d}_A) \\
&= \mathrm{d}(g) \circ f + (-1)^m g \circ \mathrm{d}(f)
\end{aligned}$$

as desired.

Example 3.9 (Graded category of graded modules)**.** Let A be a $\mathbf{Z}$-graded algebra over a ring R. We will construct a graded category Mod_A^{gr} over R whose associated category $(\mathrm{Mod}_A^{gr})^0$ is the category of graded A-modules. As objects of Mod_A^{gr} we take right graded A-modules (see Section 4). Given graded A-modules L and M we set

$$\mathrm{Hom}_{\mathrm{Mod}_A^{gr}}(L, M) = \bigoplus\nolimits_{n \in \mathbf{Z}} \mathrm{Hom}^n(L, M)$$

where $\mathrm{Hom}^n(L, M)$ is the set of right A-module maps $L \to M$ which are homogeneous of degree n, i.e., $f(L^i) \subset M^{i+n}$ for all $i \in \mathbf{Z}$. In terms of components, we have that

$$\mathrm{Hom}^n(L, M) \subset \prod\nolimits_{p+q=n} \mathrm{Hom}_R(L^{-q}, M^p)$$

(observe reversal of indices) is the subset consisting of those $f = (f_{p,q})$ such that

$$f_{p,q}(ma) = f_{p-i,q+i}(m)a$$

for $a \in A^i$ and $m \in L^{-q-i}$. For graded A-modules K, L, M we define composition in Mod_A^{gr} via the maps

$$\mathrm{Hom}^m(L, M) \times \mathrm{Hom}^n(K, L) \longrightarrow \mathrm{Hom}^{n+m}(K, M)$$

by simple composition of right A-module maps: $(g, f) \mapsto g \circ f$.

Lemma 3.10. *Let (A, d) be a differential graded algebra. Let I be a differential graded A-module. If $F_\bullet$ is a filtration as in property (I), then we obtain an admissible short exact sequence*

$$0 \to I \to \prod I/F_i I \to \prod I/F_i I \to 0$$

of differential graded A-modules.

Proof. Omitted. Hint: This is dual to Lemma 3.13. □

Proposition 3.11. *Let (A, d) be a differential graded algebra. The homotopy category $K(Mod_{(A,d)})$ of differential graded A-modules with its natural translation functors and distinguished triangles is a triangulated category.*

Proof. We know that $K(\mathrm{Mod}_{(A,\mathrm{d})})$ is a pre-triangulated category. Hence it suffices to prove TR4 and to prove it we can use Derived Categories, Lemma 014C. Let $K \to L$ and $L \to M$ be composable morphisms of $K(\mathrm{Mod}_{(A,\mathrm{d})})$. By Lemma 4.1 we may assume that $K \to L$ and $L \to M$ are admissible monomorphisms. In this case the result follows from Lemma 4.2. □

Lemma 3.12. *Let (A, d) be a differential graded algebra. Then we have*

$$\mathrm{Hom}_{Mod_{(A,d)}}(A[k], M) = \mathrm{Ker}(d : M^{-k} \to M^{-k+1})$$

and

$$\mathrm{Hom}_{K(Mod_{(A,d)})}(A[k], M) = H^{-k}(M)$$

for any differential graded A-module M.

Proof. Immediate from the definitions. □

Lemma 3.13. *Let (A, d) be a differential graded algebra. Let P be a differential graded A-module. If $F_\bullet$ is a filtration as in property (P), then we obtain an admissible short exact sequence*

$$0 \to \bigoplus F_i P \to \bigoplus F_i P \to P \to 0$$

of differential graded A-modules.

Proof. The second map is the direct sum of the inclusion maps. The first map on the summand F_iP of the source is the sum of the identity $F_iP \to F_iP$ and the negative of the inclusion map $F_iP \to F_{i+1}P$. Choose homomorphisms $s_i : F_{i+1}P \to F_iP$ of graded A-modules which are left inverse to the inclusion maps. Composing gives maps $s_{j,i} : F_jP \to F_iP$ for all $j > i$. Then a left inverse of the first arrow maps $x \in F_jP$ to $(s_{j,0}(x), s_{j,1}(x), \ldots, s_{j,j-1}(x), 0, \ldots)$ in $\bigoplus F_iP$. □

Lemma 3.14. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $M \to I$ of differential graded A-modules with the following properties*

1. *$M \to I$ is injective,*
2. *$\mathrm{Coker}(d_M) \to \mathrm{Coker}(d_I)$ is injective, and*
3. *I sits in an admissible short exact sequence $0 \to I' \to I \to I'' \to 0$ where I', I'' are products of shifts of $A^\vee$.*

Proof. We will use the functors $N \mapsto N^\vee$ (from left to right differential graded modules and from right to left differential graded modules) constructed in Section 5 and all of their properties. For every $k \in \mathbf{Z}$ let Q_k be the free left A-module with generators x, y in degrees k and $k+1$. Define the structure of a left differential graded A-module on Q_k by setting $\mathrm{d}(x) = y$ and $\mathrm{d}(y) = 0$. Arguing exactly as in the proof of Lemma 3.5 we find a surjection

$$\bigoplus\nolimits_{i \in I} Q_{k_i} \longrightarrow M^\vee$$

of left differential graded A-modules. Then we can consider the injection

$$M \to (M^\vee)^\vee \to (\bigoplus\nolimits_{i \in I} Q_{k_i})^\vee = \prod\nolimits_{i \in I} I_{k_i}$$

where $I_k = Q_{-k}^\vee$ is the "dual" right differential graded A-module. Further, the short exact sequence $0 \to A[-k-1] \to Q_k \to A[-k] \to 0$ produces a short exact sequence $0 \to A^\vee[k] \to I_k \to A^\vee[k+1] \to 0$.

The result of the previous paragraph produces $M \to I$ having properties (1) and (3). To obtain property (2), suppose $\overline{m} \in \mathrm{Coker}(\mathrm{d}_M)$ is a nonzero element of degree k. Pick a map $\lambda : M^k \to \mathbf{Q}/\mathbf{Z}$ which vanishes on $\mathrm{Im}(M^{k-1} \to M^k)$ but not on m. By Lemma 1.5 this corresponds to a homomorphism $M \to A^\vee[k]$ of differential graded A-modules which does not vanish on m. Hence we can achieve (2) by adding a product of copies of shifts of $A^\vee$. □

Lemma 3.15. *Let (A, d) be a differential graded algebra. Let $I \to M$ be an injective homomorphism of differential graded A-modules. If I is graded injective, then $I \to M$ is an admissible monomorphism.*

Proof. This is immediate from the definitions. □

Example 3.16 (Graded category of graded objects)**.** Let $\mathcal{B}$ be an additive category. Recall that we have defined the category $\mathrm{Gr}(\mathcal{B})$ of graded objects of $\mathcal{B}$ in Homology, Definition 0125. In this example, we will construct a graded category $\mathrm{Gr}^{gr}(\mathcal{B})$ over $R = \mathbf{Z}$ whose associated category $\mathrm{Gr}^{gr}(\mathcal{B})^0$ recovers $\mathrm{Gr}(\mathcal{B})$. As objects of $\mathrm{Gr}^{gr}(\mathcal{B})$ we take graded objects of $\mathcal{B}$. Then, given graded objects $A = (A^i)$ and $B = (B^i)$ of $\mathcal{B}$ we set

$$\mathrm{Hom}_{\mathrm{Gr}^{gr}(\mathcal{B})}(A, B) = \bigoplus\nolimits_{n \in \mathbf{Z}} \mathrm{Hom}^n(A, B)$$

where the graded piece of degree n is the abelian group of homogeneous maps of degree n from A to B. Explicitly we have

$$\mathrm{Hom}^n(A, B) = \prod\nolimits_{p+q=n} \mathrm{Hom}_{\mathcal{B}}(A^{-q}, B^p)$$

(observe reversal of indices and observe that we have a product here and not a direct sum). In other words, a degree n morphism f from A to B can be seen as a system $f = (f_{p,q})$ where $p, q \in \mathbf{Z}$, $p + q = n$ with $f_{p,q} : A^{-q} \to B^p$ a morphism of $\mathcal{B}$. Given graded objects A, B, C of $\mathcal{B}$ composition of morphisms in $\mathrm{Gr}^{gr}(\mathcal{B})$ is defined via the maps

$$\mathrm{Hom}^m(B, C) \times \mathrm{Hom}^n(A, B) \longrightarrow \mathrm{Hom}^{n+m}(A, C)$$

by simple composition $(g, f) \mapsto g \circ f$ of homogeneous maps of graded objects. In terms of components we have

$$(g \circ f)_{p,r} = g_{p,q} \circ f_{-q,r}$$

where q is such that $p + q = m$ and $-q + r = n$.

4. Projective modules over algebras

Lemma 4.1. *Let (A, d) be a differential graded algebra. Let $L_1 \to L_2 \to \ldots \to L_n$ be a sequence of composable homomorphisms of differential graded A-modules. There exists a commutative diagram*

$$\begin{array}{ccccccc} L_1 & \longrightarrow & L_2 & \longrightarrow & \ldots & \longrightarrow & L_n \\ \uparrow & & \uparrow & & & & \uparrow \\ M_1 & \longrightarrow & M_2 & \longrightarrow & \ldots & \longrightarrow & M_n \end{array}$$

in $Mod_{(A,d)}$ such that each $M_i \to M_{i+1}$ is an admissible monomorphism and each $M_i \to L_i$ is a homotopy equivalence.

Proof. The case $n = 1$ is without content. Lemma 5.1 is the case $n = 2$. Suppose we have constructed the diagram except for M_n. Apply Lemma 5.1 to the composition $M_{n-1} \to L_{n-1} \to L_n$. The result is a factorization $M_{n-1} \to M_n \to L_n$ as desired. □

Lemma 4.2. *Let (A, d) be a differential graded algebra. Suppose that $\alpha : K \to L$ and $\beta : L \to M$ are admissible monomorphisms of differential graded A-modules. Then there exist distinguished triangles $(K, L, Q_1, \alpha, p_1, d_1)$, $(K, M, Q_2, \beta \circ \alpha, p_2, d_2)$ and $(L, M, Q_3, \beta, p_3, d_3)$ for which TR4 holds.*

Proof. Say $\pi_1 : L \to K$ and $\pi_3 : M \to L$ are homomorphisms of graded A-modules which are left inverse to α and β. Then also $K \to M$ is an admissible monomorphism with left inverse $\pi_2 = \pi_1 \circ \pi_3$. Let us write Q_1, Q_2 and Q_3 for the cokernels of $K \to L$, $K \to M$, and $L \to M$. Then we obtain identifications (as graded A-modules) $Q_1 = \mathrm{Ker}(\pi_1)$, $Q_3 = \mathrm{Ker}(\pi_3)$ and $Q_2 = \mathrm{Ker}(\pi_2)$. Then $L = K \oplus Q_1$ and $M = L \oplus Q_3$ as graded A-modules. This implies $M = K \oplus Q_1 \oplus Q_3$. Note that $\pi_2 = \pi_1 \circ \pi_3$ is zero on both Q_1 and Q_3. Hence $Q_2 = Q_1 \oplus Q_3$. Consider

the commutative diagram

$$\begin{array}{ccccccccc} 0 & \to & K & \to & L & \to & Q_1 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & K & \to & M & \to & Q_2 & \to & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \to & L & \to & M & \to & Q_3 & \to & 0 \end{array}$$

The rows of this diagram are admissible short exact sequences, and hence determine distinguished triangles by definition. Moreover downward arrows in the diagram above are compatible with the chosen splittings and hence define morphisms of triangles

$$(K \to L \to Q_1 \to K[1]) \longrightarrow (K \to M \to Q_2 \to K[1])$$

and

$$(K \to M \to Q_2 \to K[1]) \longrightarrow (L \to M \to Q_3 \to L[1]).$$

Note that the splittings $Q_3 \to M$ of the bottom sequence in the diagram provides a splitting for the split sequence $0 \to Q_1 \to Q_2 \to Q_3 \to 0$ upon composing with $M \to Q_2$. It follows easily from this that the morphism $\delta : Q_3 \to Q_1[1]$ in the corresponding distinguished triangle

$$(Q_1 \to Q_2 \to Q_3 \to Q_1[1])$$

is equal to the composition $Q_3 \to L[1] \to Q_1[1]$. Hence we get a structure as in the conclusion of axiom TR4. □

5. Injective modules and differential graded algebras

Lemma 5.1. *Let (A, d) be a differential graded algebra. Let $\alpha : K \to L$ be a homomorphism of differential graded A-modules. There exists a factorization*

$$K \xrightarrow{\tilde{\alpha}} \tilde{L} \xrightarrow{\pi} L, \quad K \xrightarrow{\alpha} L$$

in $Mod_{(A,d)}$ such that

1. *$\tilde{\alpha}$ is an admissible monomorphism (see Definition 5.2),*
2. *there is a morphism $s : L \to \tilde{L}$ such that $\pi \circ s = id_L$ and such that $s \circ \pi$ is homotopic to $id_{\tilde{L}}$.*

Proof. The proof is identical to the proof of Derived Categories, Lemma 013N. Namely, we set $\tilde{L} = L \oplus C(1_K)$ and we use elementary properties of the cone construction. □

Definition 5.2. Let (A, d) be a differential graded algebra.

1. A homomorphism $K \to L$ of differential graded A-modules is an *admissible monomorphism* if there exists a graded A-module map $L \to K$ which is left inverse to $K \to L$.
2. A homomorphism $L \to M$ of differential graded A-modules is an *admissible epimorphism* if there exists a graded A-module map $M \to L$ which is right inverse to $L \to M$.
3. A short exact sequence $0 \to K \to L \to M \to 0$ of differential graded A-modules is an *admissible short exact sequence* if it is split as a sequence of graded A-modules.

Lemma 5.3. *Let R be a ring. Let $\mathcal{C}$ be a site. Let $\mathcal{O}$ be a sheaf of commutative R-algebras. Let $K^\bullet$ be a complex of $\mathcal{O}$-modules. The functor of Lemma 0.1 has the following property: For every M, N in $D(E, d)$ there is a canonical map*

$$R\operatorname{Hom}(M, N) \longrightarrow R\operatorname{Hom}_{\mathcal{O}}(M \otimes_E^{\mathbf{L}} K^\bullet, N \otimes_E^{\mathbf{L}} K^\bullet)$$

in $D(R)$ which on cohomology modules gives the maps

$$\operatorname{Ext}^n_{D(E,d)}(M, N) \to \operatorname{Ext}^n_{D(\mathcal{O})}(M \otimes_E^{\mathbf{L}} K^\bullet, N \otimes_E^{\mathbf{L}} K^\bullet)$$

induced by the functor $- \otimes_E^{\mathbf{L}} K^\bullet$.

Proof. The right hand side of the arrow is the global derived hom introduced in Cohomology on Sites, Section 0B6E which has the correct cohomology modules. For the left hand side we think of M as a (R, A)-bimodule and we have the derived Hom introduced in Section 1 which also has the correct cohomology modules. To prove the lemma we may assume M and N are differential graded E-modules with property (P); this does not change the left hand side of the arrow by Lemma 0.2. By Lemma 0.3 this means that the left hand side of the arrow becomes $\operatorname{Hom}_{\operatorname{Mod}^{dg}_{(B,\mathrm{d})}}(M, N)$. In Lemmas 0.4, 0.5, and 0.1 we have constructed a functor

$$- \otimes_E K^\bullet : \operatorname{Mod}^{dg}_{(E,\mathrm{d})} \longrightarrow \operatorname{Comp}^{dg}(\mathcal{O})$$

of differential graded categories and we have shown that $- \otimes_E^{\mathbf{L}} K^\bullet$ is computed by evaluating this functor on differential graded E-modules with property (P). Hence we obtain a map of complexes of R-modules

$$\operatorname{Hom}_{\operatorname{Mod}^{dg}_{(B,\mathrm{d})}}(M, N) \longrightarrow \operatorname{Hom}_{\operatorname{Comp}^{dg}(\mathcal{O})}(M \otimes_E K^\bullet, N \otimes_E K^\bullet)$$

For any complexes of $\mathcal{O}$-modules $\mathcal{F}^\bullet$, $\mathcal{G}^\bullet$ there is a canonical map

$$\operatorname{Hom}_{\operatorname{Comp}^{dg}(\mathcal{O})}(\mathcal{F}^\bullet, \mathcal{G}^\bullet) = \Gamma(\mathcal{C}, \mathcal{H}om^\bullet(\mathcal{F}^\bullet, \mathcal{G}^\bullet)) \longrightarrow R\operatorname{Hom}_{\mathcal{O}}(\mathcal{F}^\bullet, \mathcal{G}^\bullet).$$

Combining these maps we obtain the desired map of the lemma. □